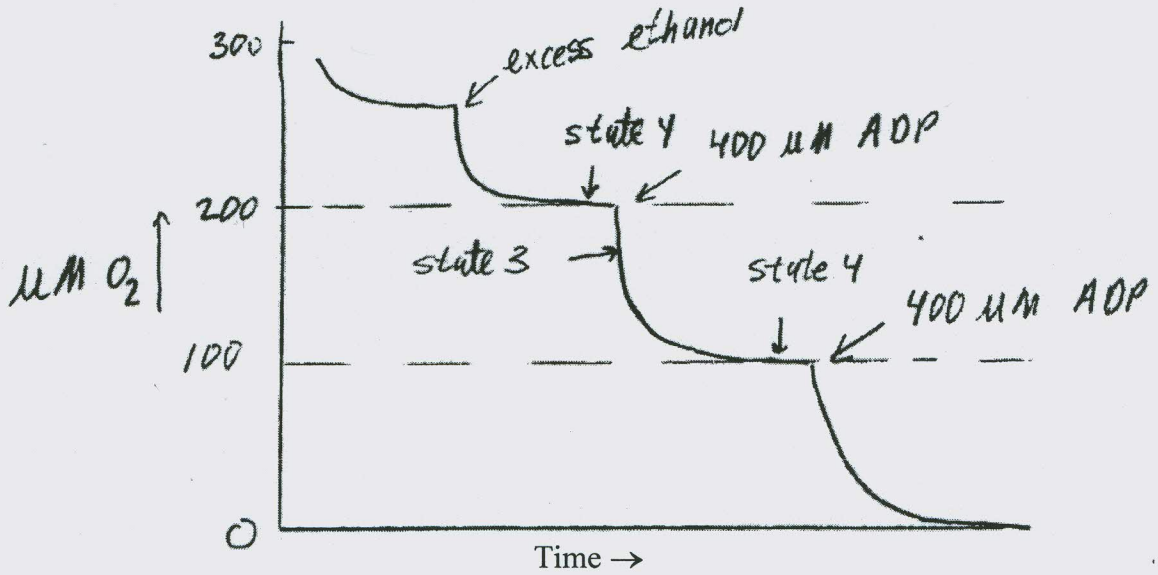
Biochemistry Cume – March 3, 2006

A new organism has been discovered that uses ethanol as an energy source and contains mitochondria that transfer electrons from ethanol to O_2 to form acetaldehyde and H_2O. (acetaldehyde + 2 e^- + 2 H^+ → ethanol; $E_o' = -0.2$ V) (½ O_2 + 2 e^- + 2 H^+ → H_2O; $E_o' = +0.82$ V). The mitochondria contain five redox centers, A, B, C, D, and E, and an ATP synthase with the same properties as in mammalian mitochondria. The following experiments were carried out on intact mitochondria from this organism:

a) O_2 consumption studies:



b) Spectroscopic studies of the percent reduction of each redox center:

	A	B	C	D	E
State 3	70%	30%	50%	10%	90%
State 4	100%	50%	50%	0%	100%

Questions:

1. (20) Calculate the total exergonic energy released during electron transport from 1 mole ethanol to O_2

2. (40) From the above studies, calculate how many moles of ATP were synthesized per mole of ethanol oxidized. Calculate the actual free energy change $\Delta G'$ for the synthesis of ATP under cellular conditions: [ATP] = 20 mM, [P_i] = 10 mM, [ADP] = 2 mM. Calculate the efficiency with which the exergonic energy of electron transfer is conserved by ATP production.

3. (40) Determine the order of the redox centers involved in electron transport from ethanol to O_2, and the sites at which electron transfer is coupled to proton pumping. Draw a model for electron transport and ATP production in these mitochondria.

Possibly useful formulas:

$\Delta G^{o'} = -RT\ln K_{eq}$; $\Delta G^o = -nF\Delta E_o$
$\Delta G = \Delta G^o + RT\ln\{\text{products/reactants}\}$
F = Faraday's constant = 23 kcal/volt; R = 1.987cal mol^{-1} deg^{-1}
ΔG^o(ATP hydrolysis) = -7.3 kcal/mol